HKU MAIB7002 | GROUP Q | FRESH RETAIL

FRESH FOOD. SMARTER DISCOUNTS.

Should we discount it tomorrow?



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Help a category manager choose tomorrow's discount.

Unsold perishables create waste; unnecessary markdowns give away value. Can forecasts help?

WHY THESE DATA?

FreshRetailNet records daily sales, discounts and stockouts. We select regularly selling products with repeated price changes.

50,000 public histories → **16,684** eligible histories → **312** in five stores

Filters: 15-85% discounted days (rate <0.95), 6+ discount switches, <20% zero-sales days. Then take the five stores with the most eligible histories.

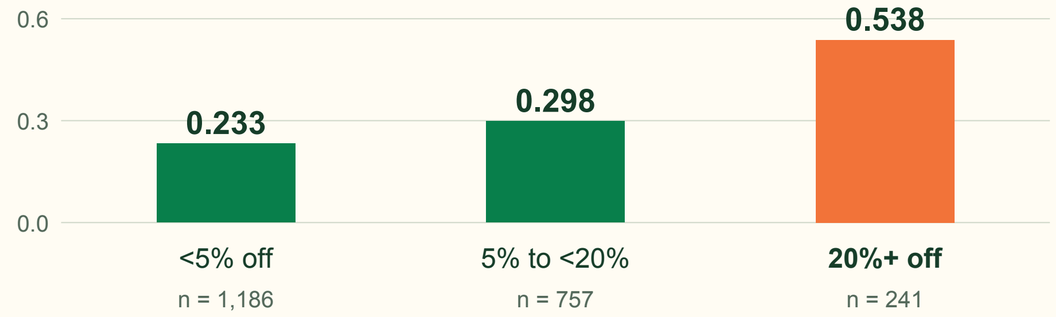
1 A better forecast - with difficult cases still to solve

Predict next-day normalized sales; choose settings with five time-ordered validation weeks.

Mean absolute error (MAE) — Lower is better



Ensemble MAE by observed discount

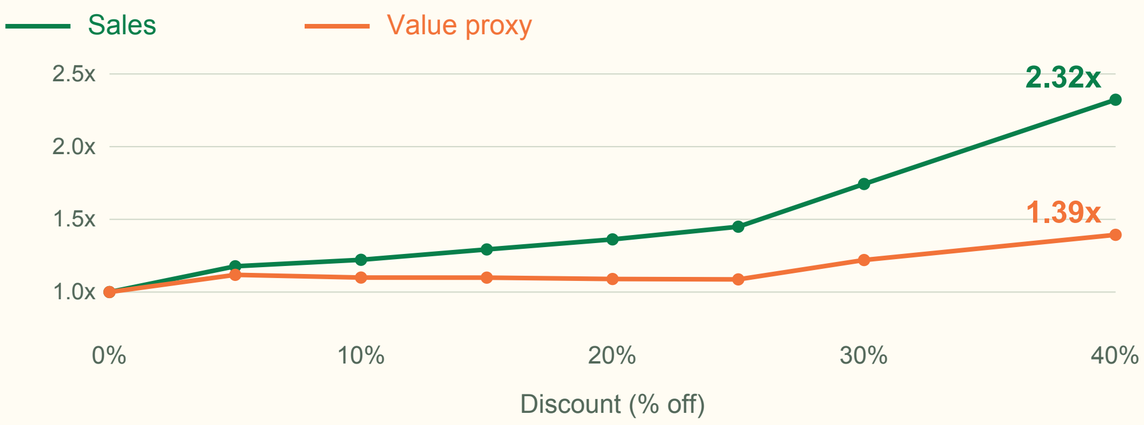


7.9% lower MAE vs. the 7-day average. Equal-weight RF + CatBoost; CatBoost alone is nearly tied. Deep-discount days have larger absolute errors and higher average sales.

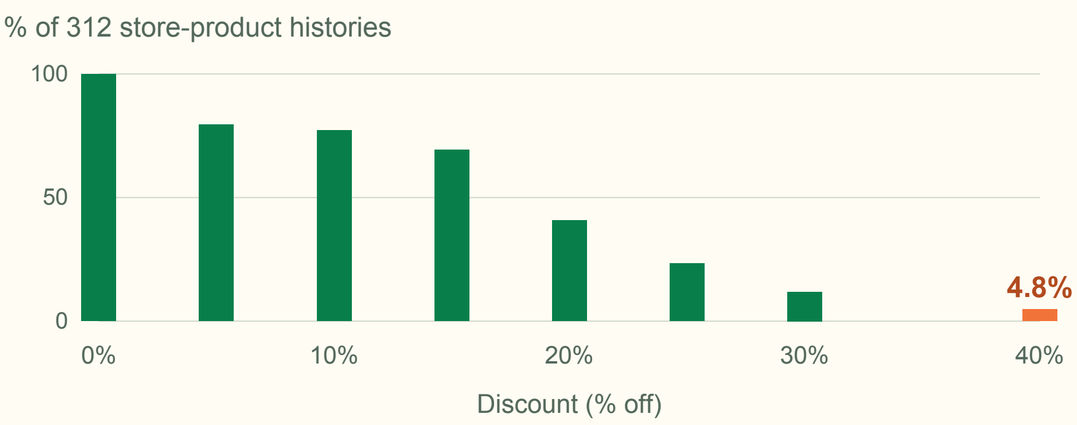
2 Bigger predicted gains, thinner historical evidence

Same 2,184 cases, eight discounts. Value proxy = predicted sales x price rate; zero cost assumed.

Predicted response vs. full price



Histories supporting each candidate



40% off 2.32x predicted sales, but only 15 / 312 histories (4.8%) support this rate. Support requires 3+ training days within 0.025 of the candidate price rate.

3 Add explicit stock rules before choosing the price

Model + support still recommends discounts almost every day; rules change when and how deeply.

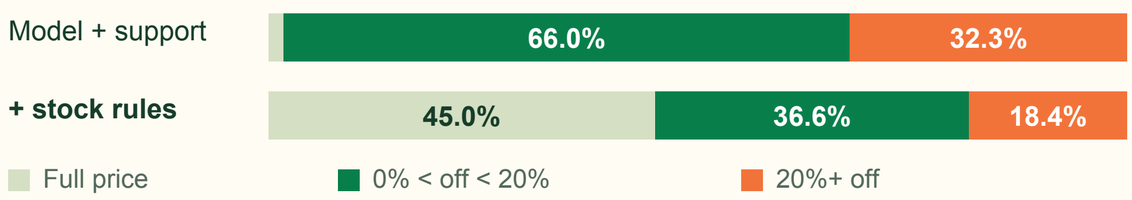
1 Stockout today >=1 hour
Full price tomorrow

2 Otherwise, no stockout in 7 days
Use a supported discount, if available

3 Otherwise
Choose the highest predicted value

Discount recommendations

98.3% → **55.0%** with stock rules



Same 2,184 cases. Rules are assumptions; stockouts do not reveal inventory age. Revenue or waste gains remain untested.

ONE PRODUCT, ONE DECISION: 5% OFF

Store 18 / Product 11 / 26 June

Predicted value: Full price 1.121

Best eligible rate: 5% off 1.198

Next: validate decisions in a controlled store pilot.

Observational, normalized sales; value is a proxy, not currency. Five stores in one city; retrospective selection. 26 Jun - 2 Jul 2024: 2,184 cases, a reused benchmark. Add inventory and cost data before a pilot.

Source: Dingdong FreshRetailNet-50K (CC BY 4.0). Final notebook: pricing-ml-ap @ a7e26dd. Model scenarios do not prove causal uplift.